

THE MEASURED COST OF CACHED-DEPTH READ-OUT: A MATCHED, PRE-REGISTERED NEGATIVE RESULT FOR HIDDEN-STATE CACHE IN LONG-CONTEXT IN- FERENCE

Anonymous authors

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ABSTRACT

Long-context systems commonly retain or reconstruct key-value state at every layer. We study a training-free alternative on a frozen backbone: cache one activation tensor per chunk at depth j , retrieve a fixed number of chunks, and recompute only the upper layers. **CoMem** implements this write-retrieve-read pipeline. With fixed top- k , model read length is independent of document length and the measured GPU read-time working set remains approximately flat through 128k tokens. The external activation store, raw text, and BM25 index still grow with the document, so this is not a constant-memory system.

Matched measurements on Qwen3-8B separate the retrieval operating point from the depth-cache trade-off. A fixed- k short read provides the working-set benefit. At the same 6.7k-token read in a 128k memory check, moving from $j=0$ to $j=12$ saves about 0.3 GB, a read-length-scoped quantity rather than a general memory benefit. The matched $j=12$ read is less accurate than re-forwarding the same retrieved chunks from $j=0$. In a five-arm decomposition, the contrast remains significant when selection, decoding, chunk-independent encoding, and position relocation are held fixed; a residual lower-band context-set difference prevents interpreting it as a fully deconfounded causal effect. The relocation contrast is null at the scored endpoint but not at the raw-generation level.

The preregistered depth sweep has the same direction: pointwise and without multiplicity correction, every tested $j \geq 2$ read loses to matched $j=0$ on the same samples, and no deeper point recovers it. The load-bearing $j=12$ comparison remains significant under Bonferroni correction, although the curve is not strictly monotone. Because the headline 4k magnitude was selected and depends on the registered reader convention, the reusable conclusion is the direction, not a convention-independent effect size. The result bounds what this frozen, training-free backbone gives up; it does not bound recovery after training for cache compressibility or establish superiority over unimplemented prior systems.

1 INTRODUCTION

Long-context inference has two different scaling problems: the model’s active working set grows with the sequence it rereads, and persistent state grows as more text is retained. KV eviction bounds the former by discarding tokens (Xiao et al., 2024b; Zhang et al., 2023; Li et al., 2024) and retrieval bounds it by selecting a short pack, but both usually retain raw tokens or full-layer KV. We ask whether the *depth* axis provides another operating point: can a system retain one intermediate activation tensor per chunk and reconstruct only the layers above it when that chunk is retrieved?

CoMem answers this question with a deliberately simple, *training-free* pipeline on a frozen backbone (Figure 1). A write pass chunks a document, runs each chunk through layers $[0, j]$ with chunk-local positions, and stores the $c \times d$ residual activation h_j together with a raw-text BM25 index. A read selects fixed top- k chunks, forms $[\text{sink}; h_j^{(1)}, \dots, h_j^{(k)}; h_j^q]$ with fresh contiguous positions,

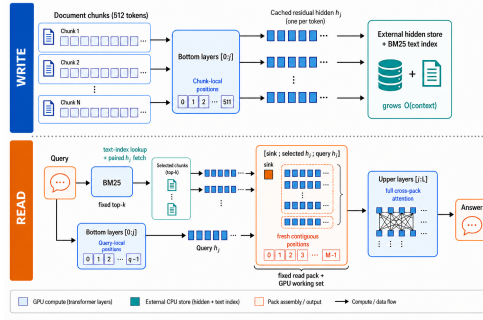


Figure 1: **CoMem write-retrieve-read path.** Each 512-token document chunk is written through layers $[0, j]$ and stored as a per-token layer- j activation with its text-index entry. BM25 selects fixed top- k chunks; their activations and the query activation form a freshly positioned pack read by layers $[j, L]$ with full cross-pack attention. The external store grows as $O(\text{context})$; the fixed- k read pack does not (its measured read-time peak stays approximately flat through 128k but is not constant).

and recomputes layers $[j, L]$ with full cross-pack attention. This is neither a fixed-size memory nor an exact transformation of the deployed input: the persistent hidden/text store is $O(\text{context})$, and relocating chunk-local $j > 0$ states changes the computation. The narrow benefit is that, for fixed k , model read length is independent of document length; the measured GPU read-time peak stays approximately flat through 128k (see Limitations).

The central experimental question is consequently not whether retrieval helps, but what the depth cache contributes *given the same retrieval*. We isolate it by comparing $j = 12$ with $j = 0$ while holding the selected chunks, query, scorer, and decoder fixed; the $j = 0$ arm re-forwards the same retrieved raw tokens through all layers and is the matched reference. That comparison exposes a directional read-out cost rather than a free compression: at the preregistered BABILong qa1 decision cell the matched $j=12$ read is worse than $j=0$ under both the frozen reader and a sensitivity reader that removes model-generated option enumerations, with the magnitude reported as an interval rather than a point value (Table 2). We therefore treat depth caching as a quality-systems trade-off, not evidence that a mid-layer state is a sufficient statistic for general “understanding.” A five-arm decomposition then freezes chunk-independent encoding and position relocation separately and together: the depth-varying arm remains worse than $j = 0$, whereas relocation is null at the scored endpoint. This isolates depth with respect to those named factors, while a residual lower-band context-set difference prevents a fully deconfounded causal interpretation (Section 5.2).

Closest work and the gap we fill. The closest prior designs are Layer-Condensed KV (Wu & Tu, 2024), which also caches a single shallow state on the layer axis but lacks top- k retrieval, and HCache (Gao et al., 2025) and KV-Direct (Qasim et al., 2026), which restore or replace state from intermediate activations without document retrieval (Section 2, Appendix A). What this comparison set does not provide is a matched, preregistered measurement that holds retrieval, selector, scorer, and decoder fixed while changing cache depth and reports the resulting quality cost. We provide that measurement: an overall downward trend with a consistent directional loss at the registered comparisons. CoMem is the vehicle on which the protocol runs rather than the contribution by itself. Our empirical positioning is internal—full context, matched $j=0$, cached $j=12$, and the author re-forward controls; a same-bed comparison to a prior system is not run and remains future work.

Contributions.

- a **controlled measurement protocol** factorizing depth from retrieval, selection, and decoding on a frozen, training-free stack, so the depth cache is evaluated as a quality-systems trade-off, not a free compression;
- a **depth-cost curve** showing a downward trend under fixed retrieval (no tested deeper point recovers $j=0$), with matched paired CIs and pointwise tests;

- a **measured resource account** separating the $O(\text{context})$ external store and index from the fixed- k model read, including a 128k H20 serving comparison and explicit quality losses;
- CoMem, the depth-partitioned write–retrieve–read mechanism that instantiates the protocol and whose layer-axis store and upper-band recomputation are the design we measure.

2 RELATED WORK AND CONTRIBUTION BOUNDARY

Token-axis methods choose *which* token states to retain (Xiao et al., 2024b; Zhang et al., 2023; Li et al., 2024; Cai et al., 2024; Lin et al., 2026), and retrieval-and-reuse serving systems keep full-layer KV for retrieved chunks (Yao et al., 2024; Jin et al., 2024; Gim et al., 2024); both leave the layer axis untouched. The three nearest neighbours act on that axis: Layer-Condensed KV shares one condensed KV computed at a shallow depth across the upper layers (Wu & Tu, 2024), HCache restores evicted state from intermediate activations without document retrieval (Gao et al., 2025), and KV-Direct replaces full-layer KV with a residual-stream read once all retained tokens are reconstructed (Qasim et al., 2026). Each of them either lets the read grow with retained tokens or omits top- k selection, and—decisively for this paper—none measures cache-*depth* fidelity under a fixed retrieval, so the quality price of reading a cached mid-depth state has never been isolated from the retrieval that makes the read short. CoMem’s boundary is accordingly the *combination* of a one-depth per-token store, external BM25 top- k selection, and full upper-band recomputation: we claim neither the individual ingredients, nor numerical exactness for relocated $j > 0$ states, nor superiority over designs we do not implement, and the author’s full-depth re-forward control is labeled as ours rather than as an implementation of any published method. Appendix A gives the per-family comparison, the mechanism-boundary table (Table 4), the position-remap and external-memory neighbours (Xiao et al., 2024a; Sun et al., 2024; Mohtashami & Jaggi, 2023; Tworowski et al., 2023; Borgeaud et al., 2022; Wu et al., 2022; Bertsch et al., 2023), and the independence statement for the 2026 preprints (Qasim et al., 2026; Gelberg et al., 2026; Lin et al., 2026).

3 CoMEM

Let a decoder-only transformer contain L layers of width d . We use half-open layer ranges: $F_{[0,j]}$ contains the bottom j layers and $F_{[j,L]}$ the remaining layers. Documents are divided into M disjoint chunks x_m of $c = 512$ tokens. The mechanism is training-free: no parameter of the frozen backbone is updated at any point.

Write. For every chunk, CoMem computes

$$h_j^{(m)} = F_{[0,j]}(x_m; 0:c-1), \quad h_j^{(m)} \in \mathbb{R}^{c \times d}, \quad (1)$$

using chunk-local positions, and writes one layer’s per-token residual tensor per chunk to an external store. It also retains the raw text needed to build a CPU BM25 index. The hidden store therefore grows as $O(Mcd)$ and the text index as $O(|x|)$: the layer axis buys a persistent-storage *rate* reduction, not a constant-size memory (Appendix B).

Retrieve and read. Given query q , BM25 (Robertson & Zaragoza, 2009) ranks raw-text chunks and returns $S_k(q)$, the fixed top- k indices. CoMem sorts those indices back into original document order, computes the query activation h_j^q , gathers the paired stored activations, assigns fresh contiguous positions, and forms

$$P_j(q) = [h_j^{\text{sink}}; h_j^{(m)} : m \in S_k(q); h_j^q]. \quad (2)$$

Layers $F_{[j,L]}$ then process the entire pack with full causal cross-pack attention. Its length is $1 + kc + |q|$ and is independent of M when k and c are fixed. Thus model read FLOPs and GPU read-time KV working set are bounded with respect to document length; write time, persistent storage, BM25 index size, and generally retrieval work are not. The sink and query construction, the selector settings and variants, the block-diagonal ablation, and the per-decoded-token upper-band rebuild that keeps the recorded 0.68 s a read/prefill rather than an end-to-end quantity are in Appendix B.

What is and is not exact. On one contiguous sequence with unchanged positions, $F_{[j,L]}(F_{[0,j]}(x)) = F(x)$ up to implementation precision; this is an algebraic split/rejoin check. The deployed $j > 0$ path is different: chunks are encoded independently, selected, relocated, and jointly read with a query. There is no exactness guarantee for that transformation, so we measure its fidelity against the same-retrieval $j = 0$ arm, where selected raw tokens are re-forwarded through the whole model; at $j = L$ no model layers remain to condition the stored representation on the query, so intermediate j is a quality-compute knob. Two factors are confounded at every $j > 0$: (i) *chunk-independent encoding* (each chunk is written with chunk-local positions, without cross-chunk attention during the write), and (ii) *position relocation* (retrieved chunks are reassigned fresh contiguous positions at read). An identical-position split/rejoin is lossless by design and so isolates neither factor; disentangling (i) from (ii) requires the registered dual-step ablation of Section 5.2, reported separately from the headline depth curve.

4 EXPERIMENTAL PROTOCOL AND CLAIM MAP

Everything in this paper is measured on a frozen, training-free stack: the 36-layer Qwen3-8B backbone (Yang et al., 2025) with chunk size $c = 512$ and CoMem split $j = 12$, in bfloat16, with no parameter update anywhere in the pipeline. The qa1 and RULER depth studies use a fixed $k = 12$; the LongEval study uses the per-task BM25 settings of Appendix Table 12 ($k \in \{4, 6\}$), so its quality gap is not a $k=12$ operating-point claim. The surviving artifacts document no clean validation split for the earlier best- k sweeps, so rather than select k on the test set we exclude those values from headline claims: our result is a single fixed-configuration test, not a claim that $k = 12$ is optimal. How the depth cost changes with k is unknown, and a matched k -sweep at the 4k decision cell remains future work.

Table 1 is the map a reviewer should read first: each headline claim, its estimand, the control that identifies that estimand, the scope in which we assert it, and where its evidence lives. We report three estimands and never merge them: the read-depth *direction* (Section 5.1), the depth *attribution* (Section 5.2), and the *serving* account (Section 5.3). The remaining protocol detail — the RULER, LongEval and read-out-invariance studies, the depth-sweep protocol, the reader taxonomy, the five-arm cross-trial audit, and the reproducibility record — is in Appendices C and H.

Matched read-depth study (direction estimand). To isolate depth from retrieval, the decisive BABILong qa1 study (Kuratov et al., 2024) uses an oracle selector, $k = 12$, and the same selected chunks, query, greedy decoder, and Qwen3-8B checkpoint for $j = 12$ and $j = 0$; each registered context bucket contains positionally paired examples under the sample counts and generation caps of Appendix Table 12. Paired differences use the two-sided exact McNemar test (McNemar, 1947), with b the $j=12$ -correct/ $j=0$ -wrong pairs and c the reverse; the frozen primary reader extracts the first task label and the sensitivity reader first strips a leading model-generated option enumeration (full reader taxonomy in Appendix C). Pairing is verified by content hash: the registered `sample_id` is the 16-hex `sha256` prefix of the full formatted model input, and re-running the preregistered content-level join on it makes every decisive cell a complete 100/100 overlap with no orphaned, duplicated, or question-mismatched context, reproducing every published cell exactly (Appendix D); a separately stored 64-hex document digest was never retained and is not claimed.

Five-arm mechanism decomposition (attribution estimand). At fixed $j = 12$, length, selector, and decoder, we compare: A, the matched $j = 0$ full re-forward; D, deployed chunk-local write plus fresh-contiguous read; R, chunk-local write with original document positions at read; C, contiguous cross-chunk write with fresh-contiguous read; and E, contiguous cross-chunk write with original document positions. Thus E freezes both chunk-independent encoding and relocation when compared with A; the remaining lower-band context-set difference is stated with the attribution result rather than treated as eliminated. Each 4k/16k arm has $n = 100$ and uses the official `babiling.metrics.compare_answers` qa1 scorer, Wilson intervals, and two-sided exact McNemar contrasts on the same content-hash key `sample_id`. Driver pins, the one-line argument repair, and the cross-trial audit of that deck are in Appendices C and H. No other experimental arm is introduced.

ID	Headline claim as stated here	Estimand	Identifying control	Supported scope	Evidence
C1	one layer- j activation per chunk; only $F_{[j,L]}$ re-forwards a fixed- k pack of length $1+kc+ q $	mechanism (no estimand)	identical-position split/rejoin algebra	training-free, frozen Qwen3-8B, $c=512$; store is $O(\text{ctx})$	§3, Eq. (1)–(2); App. B
C2	cached depth is an axis whose price must be measured, not assumed	framing only	the matched design of the rows below	no empirical claim attaches to this row	§1, §4
C3	the fixed- k short read keeps needle-style recall near-perfect	accuracy at a fixed read	shared backbone/scorer vs. full context	niah.single serving sweep, $n=50/\text{cell}$; not every matched RULER cell	App. Tab. 7; App. Fig. 3, Tab. 11
C4	multi-hop composition (qa2) is <i>not evaluable</i> under the registered adjudicator	registered non-result	floor gate and negative controls	no length-robust qa2 claim anywhere	App. Tab. 9
C5	a self-distilled read adapter is archived, not evaluated	none	—	configuration only; no accuracy claim here	App. Tab. 14
C6	at fixed k the read length and GPU read-time working set are bounded while the store grows	serving working set	matched read length; median-of-3 CUDA peaks	one H20 through 128k; <i>nearly</i> flat, not constant	App. Tab. 7; App. Fig. 3
C7	depth buys 0.3 GB at a 6,657-token read; read/prefill break-even near 26 reads/document	serving, Q^*	same read length and hardware; decode excluded	read/prefill only; no end-to-end latency claim	App. Tab. 7; §5.3
C8	cached $j=12$ reads are <i>less accurate</i> than re-forwarding the same chunks from $j=0$	read-depth direction	oracle selector, same chunks, query and decoder; paired exact McNemar on a content-hash-verified subset	qa1 single-fact, $n=100/\text{cell}$, two caps; magnitude convention-dependent	Tab. 2; App. Tab. 5; App. Fig. 2
C8-D	the loss remains when the named encoding factors are frozen	depth attribution	five-arm A/D/R/C/E at fixed $j=12$; E-A varies cache depth with an acknowledged lower-band context-set boundary	decomposition on different samples from C8; not a replacement magnitude	Tab. 3; §5.2

Table 1: **Headline claims, estimands, identifying controls, scope, and evidence.** C2 and C5 are listed precisely because they carry *no* quantitative claim and C4 records a registered non-result, so a reader can check that nothing in the paper silently upgrades them. The three estimands (C6/C7 serving, C8 direction, C8-D attribution) come from different registered runs and are never pooled.

Serving study (serving estimand). The systems measurements use the same Qwen3-8B backbone on one H20 with 97.8 GB available memory, and all GPU peaks and prefill measurements are median-of-three. We distinguish (i) full-context rereading, (ii) retrieved raw-token recomputation at $j = 0$, and (iii) CoMem at $j = 12$. The registered record reports write, per-query read, GPU peak, and RULER/LongEval quality, but no end-to-end latency including decode, so the recorded read/prefill endpoints are tabulated as endpoints and never interpolated. We therefore make the framing explicit: the system’s benefit is a *retrieval* benefit (a fixed- k short read instead of a full-context read)—an operating point a matched $j=0$ short read also delivers without any depth cache—and depth caching is measured as a *quality* cost, not a saving. The matched $j = 0$ versus $j = 12$ direction remains the load-bearing intervention; the depth curve locates $j=12$ within an overall downward trend (no tested deeper working point recovers accuracy), without asserting a strictly monotone curve.

5 RESULTS

The results are organised by the three estimands of Table 1 and are never pooled across them: the read-depth *direction* (Section 5.1), the depth *attribution* (Section 5.2), and the *serving* account (Section 5.3).

5.1 DIRECTION ESTIMAND: THE CACHED READ-OUT IS NOT EXACT

Under the same retrieved chunks, reading from cached $j=12$ states is less accurate than re-forwarding those chunks from $j=0$. This qa1 intervention is the direct depth test, and Table 2 is its single canonical view: the mnt160 context sweep, the second registered cap (mnt512), and the pairing-free floor for the decision cell in one table. The load-bearing result is the direction $j=0 > j=12$ at both registered caps, and the two cap-specific estimates are not a single interval. Ev-

root / cell	reader	$j=12$	$j=0$	re-fwd.	Δ_{j0-j12}	exact p (b/c)
mmt160, 0k control	pinned	96%	98%	98%	+2 pp	0.50 (0/2)
mmt160, 1k	pinned	76%	84%	84%	+8 pp	0.115 (6/14)
mmt160, 2k	pinned	62%	77%	81%	+15 pp	5.9×10^{-3} (6/21)
mmt160, 4k (decision)	pinned	48%	75%	73%	+27 pp	4.6×10^{-7} (2/29)
mmt160, 4k	stripped	40%	75%	73%	+35 pp	7.9×10^{-8} (5/40)
mmt160, 8k	pinned	57%	75%	79%	+18 pp	5.3×10^{-4} (4/22)
mmt512, 4k (decision)	pinned	52%	76%	—	+24 pp	8.4×10^{-6} (3/27)
mmt512, 4k	stripped	57%	80%	—	+23 pp	1.17×10^{-4} (6/29)
<i>Same decision cell with no pairing assumption (two independent binomial arms)</i>						
mmt160, 4k (decision)	pinned	48%	75%	—	+27 pp	8.7×10^{-5} ($z=3.92$)

Table 2: **Canonical matched read-depth gap: one view of the decisive cells.** qal, oracle selector, $k=12$, frozen Qwen3-8B, $n=100$ per cell/arm; 0k is a no-distractor empty-context control. All cells are computed on the `sample_id` content-hash-verified pairing subset (100/100 contexts shared per cell, no mismatched or orphaned context; Appendix D), and every recomputed value matches the published one. Exact McNemar p is a deterministic function of (b, c), so the printed p and b/c are mutually consistent. The two **decision** rows are load-bearing; their exact conditional paired 95% CIs are $[+17.7, +30.5]$ pp (mmt160) and $[+14.1, +28.7]$ pp (mmt512), both excluding zero (Appendix J.1). The last row re-estimates that cell with *no* pairing assumption (pooled $z=3.924$; square-and-add interval at zero pairing correlation $[+13.6, +39.1]$ pp): the paired test is the efficient estimator, the unpaired result the assumption-free bound, and both exclude zero. *re-fwd.* is the authors’ full-depth re-forward without retrieval, *not* an implementation of the published KV-Direct method (Qasim et al., 2026); mmt160 and mmt512 reuse the same 100 samples at caps 160 and 512 and are not independent replications. Blank cells, the convention matrix and the answer-anchored null are in Appendix G.

ery cell is computed on the content-hash-verified paired subset — the two arms share all 100 contexts with no orphaned, duplicated, or question-mismatched context (Appendix D) — and the recompute reproduces each published value exactly. The direction holds in every displayed convention cell and across the overlapping half-split audit, which is resampling stability, not independent replication; the full six-cell convention matrix, with the missing-excluded rows and the answer-anchored null cell, is preserved in Appendix Table 8.

Three caveats bound how much of this may be reused, and all three are kept rather than resolved. First, generation-cap truncation does not explain the direction — the better readout is cut more often, so truncation would compress rather than create the gap — but the effect *size* is convention-dependent, both endpoints of the conditional range use the stripped reader, and a wider independent strip rule can reverse the sign, so no convention-independent magnitude is load-bearing. Second, the +27.0 pp 4k value is a *selected magnitude* rather than a pre-specified effect size: the preregistered anchor moved from 1k to 4k before the decision cell was inspected, and 4k is the largest significant gap among the valid displayed 2k/4k/8k cells. Third, the gap is not monotone in context length (Appendix Figure 4), and the answer-anchored sensitivity preserves the sign on the same samples while remaining underpowered at the longer cap. The supporting diagnostics (Appendix Tables 10 and 13) sharpen but do not change the pinned direction and pair.

Depth-partition curve. A preregistered depth sweep provides a complete curve of cached depth versus accuracy under the pinned reader (Appendix Table 5, Appendix Figure 2). Accuracy follows an overall downward trend in cached depth, and, pointwise without multiplicity correction, every tested point from $j=2$ onward is a significant paired loss against $j=0$ on the same samples (we claim no strictly monotone curve and run no ordered-trend test). The 16k dense and full-depth re-forward (no retrieval) baselines show no significant difference from $j=0$ at $n=100$; this does not establish equivalence but is consistent with a depth-carried read-out cost within fixed retrieval. The system benefit is the fixed- k short read, not depth: the same runs produce the quality and the wall-clock numbers on matched operation points, and that run-level wall clock does not support a *time* saving from depth. Per-depth memory peaks were not recorded (Appendix J); the

ctx	A: $j=0$	D: deployed	R: reloc. only	C: chunk-indep.	E: depth- varying	$E - A$	exact p
4k	.59 [.49,.68]	.16 [.10,.24]	.16 [.10,.24]	.31 [.23,.41]	.31 [.23,.41]	-.28	3×10^{-5}
16k	.52 [.42,.62]	.14 [.09,.22]	.14 [.09,.22]	.16 [.10,.24]	.16 [.10,.24]	-.36	$< 10^{-16}$

Table 3: **E2 five-arm depth-attribution decomposition at fixed $j = 12$.** Accuracy with 95% Wilson intervals, $n = 100$ per length/arm, scored by the official `babylong.metrics.compare_answers`; arms A/D/R/C/E are defined in Section 4. Per arm the five-stage graph `WRITE→SELECT→RELOC→FORWARD→SCORE` uses cache depth j (stored h_{12} residual versus raw re-forward) as its registered change object relative to A, with selector index, relocation rule, scorer, and paired inputs frozen (Appendix D). The uncontrolled lower-band context set remains a stated attribution boundary rather than an eliminated factor. The primary E–A test is paired exact McNemar. For D–R and C–E, $b=0$, $c=0$ and hence *discordant*=0 at both lengths — an endpoint result, not an identity of the forward computations (E–C raw generations differ on 66/100 and 59/100 items). D–C is $p = 2.7 \times 10^{-4}$ at 4k and $p = 0.77441$ at 16k; code snapshots and cross-trial provenance are in Appendix H. The 16k E–A test serialized as 0.0 at double precision and is reported as $< 10^{-16}$: a finite-precision output, not a mathematical zero.

separate E1 check measures a 0.3 GB read-length-scoped depth difference at 128k but yields neither a memory curve nor an end-to-end saving.

5.2 ATTRIBUTION ESTIMAND: FIVE-ARM DEPTH DECOMPOSITION

Table 3 freezes both write-position factors while varying depth. Its primary contrast has a single change object: between arm E and matched arm A we hold the selector index (the same BM25 chunks in the same order), the relocation rule, the scorer, and the scored items — the content-hash-verified paired subset — fixed, and vary only the cache depth at which the read pack is materialized, $j=0$ raw re-forward versus $j=12$ stored-residual re-forward. Chunk-local write independence and position relocation are inherited from the deployed arms and held constant, so they are not the change object. E is lower than A by 0.28 at 4k ($p = 3 \times 10^{-5}$) and 0.36 at 16k: depth itself costs a significant increment even when both encoding factors are frozen.

The secondary controls localize that result, and one of them is a bounded null rather than a zero. Relocation shows no discordant observations on the qa1 endpoint: the paired D–R and C–E contrasts are both 0/100 discordant, giving an exact one-sided 95% upper bound of 2.95% ($n=100$) on the endpoint discordance rate, so an endpoint effect up to roughly 3pp is not excluded. This is a bounded endpoint null at $n=100$, not a measured zero effect and not a computational identity: at the generation layer E and C are distinguishable — their raw outputs differ on 66/100 items at 4k and 59/100 at 16k (D vs R: 65/100 and 72/100) — and the 0/100 agreement appears only after the scorer collapses those differences onto a label. The null we claim is exactly this resolver-resolution statement, not that the ablation does nothing. A is best at both lengths and the depth-varying arm captures most of the fidelity loss, but the depth loss is an arithmetic mirror of the A-vs-C contrast (same discordant cells, signs reversed) that the endpoint null forces: arm E serves identification, not a fresh effect estimate. These runs use different samples and configuration from the pinned +27 pp cell, so their magnitudes are a mechanism decomposition, not replacements for that result.

The residual factor we do not control. Freezing encoding and relocation still leaves one difference between A and E that the deck cannot remove: the two arms differ in what their *lower* band attended to. A’s bottom layers see only the selected pack, whereas E’s stored h_{12} was written over the contiguous document region, so the lower-band context set is a third residual factor. E–A therefore varies depth in the registered computation graph but is not a fully deconfounded causal contrast. The registered bound on that class of retrieval-side difference is the 16k *dense/re-forward* controls, within 6 pp of $j=0$ (Appendix Table 5) and well below the 28 pp E–A effect at 4k, so a residual context-set effect of that size cannot account for the attribution. A confound-free arm that rewrites A’s lower band over the same region is a forward pass we have not run.

5.3 SERVING ESTIMAND: WHAT THE FIXED READ COSTS AND BUYS

Across the separate `niah_single` serving sweep, fixed-read CoMem retains perfect accuracy, whereas the full-context baseline’s longest point was run *outside* this Qwen3 checkpoint’s enabled positional regime; that point is greyed and annotated in Appendix Figure 3 rather than treated as a comparison, so its 128k cliff is a positional-regime artifact, not evidence of systems-level full-attention collapse. At the longest context CoMem uses $3.9\times$ less GPU memory and its registered per-query read remains approximately length-invariant, while full attention stays feasible below the card’s 97.8 GB capacity rather than failing out of memory. This memory contrast is *read/prefill-only* — decode rebuilds the upper pack at each token, so it is not an end-to-end win, and a matched $j=0$ short read (Section 5.2) delivers the same operating point. The recorded read/prefill endpoints are tabulated in Appendix Table 7 rather than plotted, because three endpoints are not a sweep; that table also prevents this comparison from being misread as a depth-cache ablation, since full context versus CoMem changes both selection and representation. Retrieval keeps the read in-window; it is a whole-system retrieval comparison, not a depth saving.

The matched E1 check instead holds the selected 6.7k-token read fixed: its median-of-three peaks are 18.2 GB at $j = 0$ and 17.9 GB at $j = 12$, reproduced in two trials (six of six measurements). The resulting 0.3 GB saving has 0.1 GB print granularity (range 0.2–0.4 GB) and matches the analytic read-time band $(j/L)B_{KV} \text{ read_len}$ with $j/L = 12/36$. We call this a *read-length-scoped analytic null*: weights and the 128k write transient dominate the 18.2 GB peak, so it is not general evidence that depth caching is memory-free. E1 is a memory endpoint and the direction estimand’s primary endpoint is qa1 accuracy, so neither predicts the other.

For one generation stream, that table’s registered write/read times imply a balanced break-even of approximately 26 repeated reads per written document ($Q^* = W/(t_{j0} - t_{j12}) = 6.65/0.26 = 25.58$ on the rounded serving endpoints). The quotient is sensitive to that rounding: the 0.26 s numerator is a difference of two median-of-three endpoints at 0.01 s print granularity, so a ± 0.01 s perturbation moves Q^* across roughly 25–27 reads/document, and we quote it as “about 26” rather than a two-decimal constant. Decode is excluded and the implementation recomputes the upper band at each generated token, so Q^* is read/prefill-only, has no established end-to-end analogue, and would shift under an accounting convention including decode. We claim no end-to-end time saving from depth; the serving-time calculation and the E1 memory check remain different protocols, and neither turns the selected 4k quality loss into a universal Pareto frontier.

Composition boundary. The multi-hop qa2 probe is not load-bearing and the archived self-distilled read adapter is not evaluated here, so neither carries a quantitative claim; both records, with the registered generation-cap sensitivity runs, are in Appendix F.

6 LIMITATIONS

The system is not constant-memory. The fixed- k pack is bounded, but GPU peak is not constant: it stays nearly flat through 128k and rises materially at the separate one-million-token serving point (Appendix H). The activation/text/index stores grow as $O(\text{context})$; index size and BM25 latency remain unmeasured.

Depth caching loses information needed by the deployed read. The same-retrieval $j = 12$ arm is worse than $j = 0$ on qa1 and the registered RULER/LongEval aggregates, and no tested deeper point recovers $j=0$ accuracy (Appendix Table 5); identical-position split/rejoin does not validate relocated chunks. The answer-anchored 512-cap result is underpowered and the directional diagnostics are post-hoc, leaving a clean magnitude open. The quality cost is also task-dependent: the fixed read retains near-perfect recall on needle-style single-supporting-fact probes, so that claim does not extend to multi-hop composition, which is not evaluable under the registered adjudicator (Appendix G). The 16k dense/re-forward controls are a non-rejection, not an equivalence test, so a retrieval-side deficit of a few pp is not excluded, though it remains far below the *depth* effect.

The result is training-free and therefore bounded. Every number here comes from a frozen backbone with no parameter update, so the measured depth cost is what a *training-free* deployment pays. It is not a statement about what is achievable in principle: representations can be trained to be

more cache-compressible (Gelberg et al., 2026), and a backbone or adapter optimized for reading relocated mid-depth states could recover part of the gap we measure. We ran no such training, so we neither claim nor exclude that recovery.

Selection, uncertainty, and scorer resolution. The archived best- k sweeps lack a validation/test separation, so we report fixed $k = 12$; primary RULER and qa1 cells have $n = 50$ and $n = 100$ under one checkpoint and seed, and overlapping half-splits diagnose resampling stability, not independent replication. The pairing join is over the formatted model input rather than an independently stored 64-hex document digest, which was never retained, so the load-bearing cell also carries the pairing-free two-proportion floor. The official scorer collapses roughly six in ten raw-generation differences onto one label, so the 0/100 endpoint agreement between arms (e.g. E and C) is indistinguishability *at label resolution*, not identity or inertness of a deleted factor (Appendix F).

Mechanism-attribution boundary of the closest prior. NOT_MEASURED: the attributed-mechanism support of the closest prior citation(s) is not verified item-by-item by a human reader; the claim of mechanism attribution to those loci (Layer-Condensed KV, HCache, KV-Direct) is therefore bounded and not asserted as human-verified.

Serving and data governance. Full attention at 128k uses 71.4 GB and remains feasible on the H20, CoMem recomputes the upper band per decoded token so 0.68 s is read/prefill rather than end-to-end latency, and the 0.3 GB E1 difference establishes no general depth-memory or end-to-end benefit. Raw text and activations require encrypted storage, access control, and source-bound deletion.

7 CONCLUSION

CoMem provides a factorized measurement of retrieval and cached depth. On the tested Qwen3-8B configurations, fixed top- k retrieval keeps the model read short as document length grows, while the external activation/text store grows linearly. At the measured 6.7k-token read, moving from $j=0$ to $j=12$ saves only 0.3 GB and reduces qa1 read-out fidelity under the registered comparisons. The five-arm study retains this loss after freezing the named encoding and relocation factors, subject to the stated lower-band context-set boundary. The reusable output is the matched protocol and its negative depth-cost curve; end-to-end value including decode, and recovery after cache-aware training, remain unmeasured.

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A EXTENDED RELATED WORK AND CONTRIBUTION BOUNDARY

Token-axis KV reduction. StreamingLLM keeps recent tokens and attention sinks (Xiao et al., 2024b); H2O keeps heavy hitters (Zhang et al., 2023); SnapKV predicts important positions from an observation window (Li et al., 2024); and PyramidKV varies the token budget across layers (Cai et al., 2024). These methods choose which token states to retain. CoMem instead retains every token of a selected chunk at only one depth, so its storage reduction is on the layer axis and does not imply that the external store is fixed-size. Layer-Condensed KV compresses the cache on the same layer axis, sharing one condensed KV computed at a shallow depth across the upper layers (Wu & Tu, 2024), but it keeps a state for every token and performs no top- k retrieval, so its read still grows with retained tokens. CoMem likewise stores at a single depth but only for selected chunks and recomputes the upper band with full cross-pack attention, bounding the read at fixed k, c . YOCO shares one global cache across a decoder-decoder architecture rather than storing retrieved per-chunk residual tensors (Sun et al., 2024). These layer-axis designs motivate the comparison but are not implemented in our harness; the paper’s empirical claims are therefore internal matched controls, not a performance ranking over prior systems.

Retrieval and cached blocks. Landmark Attention and Focused Transformer learn mechanisms for selecting distant content (Mohtashami & Jaggi, 2023; Tworowski et al., 2023). Serving systems such as CacheBlend and RAGCache reuse KV for retrieved chunks and selectively repair cross-chunk interactions (Yao et al., 2024; Jin et al., 2024). CoMem shares their retrieval-first view but stores a layer- j activation rather than full-layer KV, then recomputes the complete upper band with full cross-pack attention. BM25 is an external, non-differentiable selector rather than a learned contribution. InFLLM keeps distant content in memory blocks that use unit/separate positions and anchors the relevant blocks into the current segment, selecting blocks by attention (Xiao et al., 2024a). CoMem shares the position-remap idea (fresh contiguous positions for retrieved layer- j activations) but selects chunks by external BM25 rather than attention, stores a single-layer activation instead of a unit-positioned block summary, and recomputes the complete upper band with full cross-pack attention rather than attending to compressed anchors. RETRO injects retrieved neighbors through deep cross-attention (Borgeaud et al., 2022); Memorizing Transformers query an external k NN memory from a single layer (Wu et al., 2022); Prompt Cache reuses modular prompt segments through a position-remapping schema (Gim et al., 2024); and Unlimiformer scales retrieval into encoder-decoder cross-attention (Bertsch et al., 2023). These are closest on retrieval or reusable state, but none couples a one-depth per-token store with fixed top- k selection and full upper-band recomputation.

Intermediate-state restoration. HCache restores evicted model state from intermediate activations, primarily as a serving mechanism without document retrieval (Gao et al., 2025). KV-Direct shows that a residual stream determines keys and values and can replace full-layer KV when all retained tokens are reconstructed (Qasim et al., 2026). CoMem combines partial-depth recomputation with top- k retrieval; unlike KV-Direct, it does not retain all tokens in the active read, and unlike HCache it explicitly selects chunks. This combination is the contribution boundary. We do not claim the individual ingredients, numerical exactness for relocated $j > 0$ states, or a system-level constant-memory store. Training representations for KV compressibility (Gelberg et al., 2026) and CompressKV’s semantic-retrieval-head-guided token eviction (Lin et al., 2026) are complementary token/representation-axis approaches. We treat Qasim et al.’s KV-Direct, Gelberg et al.’s representation-training study, and Lin et al.’s CompressKV as 2026 arXiv preprints (Qasim et al., 2026; Gelberg et al., 2026; Lin et al., 2026); CoMem was developed independently, and its contribution boundary does not rely on any one of them.

Direct answers to the nearest-neighbor questions. Which proposal is nearest? Layer-Condensed KV (Wu & Tu, 2024) shares the core idea of caching at a single shallow depth; HCache (Gao et al., 2025) and KV-Direct (Qasim et al., 2026) share the restoration-by-intermediate-state idea. **What does it lack?** All three either let the read grow with retained tokens (Layer-Condensed KV, and reconstruction-based KV-Direct when all tokens are retained) or omit top- k selection (HCache), and none measures cache-depth fidelity under a fixed retrieval. **Which falsifiable gap do we fill?** A matched, preregistered measurement that fixes retrieval and varies only cached depth, exposing that the depth cache—not retrieval—carries the read-out quality loss. This

Method	Persistent object	Selection	Model read as context grows	Recomputation
KV-Direct (Qasim et al., 2026)	residual per retained token	none	grows with retained tokens	full-depth reconstruction
HCache (Gao et al., 2025)	intermediate activation	none	grows with restored chunks	upper-state restoration
YOCO (Sun et al., 2024)	one global decoder cache	none	grows with retained tokens	shared cross-decoder cache
CacheBlend (Yao et al., 2024)	full KV for retrieved chunks	retrieval	fixed at fixed retrieval budget	selective repair
Layer-Condensed KV (Wu & Tu, 2024)	single condensed KV at shallow depth	none	grows with retained tokens	relax upper layers
InfLLM (Xiao et al., 2024a)	memory blocks (unit/separate positions)	attention	anchors selected blocks	attend to compressed anchors
CoMem	one layer's $c \times d$ residual tensor per chunk plus text index	BM25 top- k	fixed at fixed k, c	full upper band

Table 4: Mechanism boundary. CoMem’s distinctive combination is top- k retrieval plus upper-band recomputation from a one-layer activation store; it does not subsume the individual retrieval, residual-caching, or restoration ideas.

is a measurement/negative-result contribution; we do not claim to outperform these designs on absolute benchmark scores, and in the body the author’s full-depth re-forward control is labeled as such, not as an implementation of any of them.

B METHOD DETAILS

Autoregressive generation boundary. The recorded 0.68 s quantity ends after read/prefill. During generation the current reference path rebuilds the upper-band pack at each decoded token; it does not reuse an upper-layer KV cache across decode steps. Generated tokens are appended after the query at fresh contiguous positions and pass through the same $F_{[j,L)}$ band. We therefore do not convert the recorded prefill endpoint into an end-to-end decode or throughput claim.

Sink and query construction in the read pack. In Eq. (2) the sink is one BOS state computed independently through $F_{[0,j)}$ and reused within the read; its lower-layer context is only that BOS token. The query is separately computed through $F_{[0,j)}$ with query-local positions and occupies the final fresh pack positions after the retrieved chunks.

Selectors and cross-chunk interaction. The primary selector is one-pass BM25 with $k_1 = 1.5$ and $b = 0.75$. An exploratory `iter_bm25` variant performs CPU-only chain following: round one uses the query, and later rounds re-query with raw tokens from the previous round’s newly selected chunks. It performs no model read between hops. Full cross-pack attention is part of the CoMem read; a block-diagonal upper-band recompute is an ablation, not the default.

Storage accounting. The hidden store scales as $O(Mcd)$ and the text index as $O(|x|)$; both grow with context. Under Qwen3-8B’s grouped-query attention, one per-token activation stores one eighteenth of the elements of the full layer-wise key-value cache. This is a persistent-storage rate reduction, not a constant-size memory. Table 7’s GPU peaks compare fixed read with full-context prefill, not depth alone; no end-to-end persistent-storage baseline arm is run.

C FULL EXPERIMENTAL PROTOCOL

RULER fixed- k study. We evaluate the RULER `niah_single` serving sweep (Hsieh et al., 2024) at 8k–128k with the official `string_match_all` metric. CoMem and the full-context reference share the backbone and scorer; their read primitives differ. We report Wilson 95% binomial intervals (Wilson, 1927); the sample count and seed are in Table 12. These cells quantify a complete short read versus a full-context read, so they reflect the combined effect of retrieval and working-set reduction, not the independent contribution of depth caching. Figure 3’s 100% statement is restricted to this `niah_single` fixed-read sweep; it is a different subtask and sample set from Table 11’s `niah_single_3`, not a claim about every matched RULER cell.

Matched RULER study. The depth-only RULER audit holds an oracle/retrieval-fixed selector constant and compares the same items under $j = 0$ and $j = 12$ across the registered family-length cells. Table 12 records the shards, pooled sample count, seed, and generation cap. Per-item `string_match_all` is fractional recall; strict correctness requires all references. Table 11 reports the cells, the separately registered pooled full-match audit and its descriptive exact test, and the serving-sweep exception, while Table 7 carries only the cell-level macro row. Fractional cell means are not used to reconstruct binary identities or the pooled tally.

Read-out invariance check. A separately preregistered higher-cap `qa1` check tests read-out invariance. The legacy terminal-punctuation control is absorbing and therefore does *not* support a load-bearing truncation diagnosis. The effective at-cap proxy cuts the better arm more often, so truncation cannot explain the direction. Table 10 preserves the cap, sample, reader, and discordant-cell details and labels the post-hoc diagnostics.

LongEval protocol. We use the released LongEval lines-retrieval benchmark (Li et al., 2023) over the registered context range. BM25 selects top- k raw-text chunks under the per-task settings in Table 12 ($k \in \{4, 6\}$, not the $k=12$ used at `qa1/RULER`); the metric is exact recovery of the first valid line identifier in the full output. Samples were generated separately for the $j = 0$ and $j = 12$ arms and are not in one-to-one correspondence, so the pooled LongEval column is *not* a same-retrieval depth isolation. We therefore use an unpaired two-proportion test, not McNemar; Table 7 reports the pooled estimate and test and is labeled accordingly.

Depth-partition sweep protocol. The preregistered depth sweep uses the same byte-pinned reader, oracle selector, and $n=100/\text{bucket/arm}$ as the decisive study (Table 5, Figure 2). At both tested context lengths, accuracy shows an overall downward trend in cached depth and, pointwise without multiplicity correction, every tested point from $j=2$ onward is a significant paired loss against $j=0$ on the same samples (the observed curve is not strictly monotone—one 1 pp adjacent reversal at $j=8 \rightarrow 10$ at 4k is within sampling noise at $n=100$; we run no ordered-trend test). The 16k dense (stock full-context, no CoMem) and full-depth re-forward (no retrieval) baselines show no significant difference from $j=0$ at $n=100$ ($p=0.43/0.50$); this does not prove equivalence, but the observed ≤ 6 pp deficit is far below the observed depth-associated ≥ 29 pp effect. The per-depth values and the same-run wall-clock are merged into Table 5; per-depth sweep memory peaks were not recorded (Appendix J), and the separate matched 128k memory check is reported in Table 7. Optional LoRA and MoE diagnostics with incomplete configurations remain in Appendix G and do not support the main claim.

Terminology used in the protocol and result tables. The *primary reader* extracts the first task label; the *stripped* sensitivity first removes a model-generated option list, and the *answer-anchored* sensitivity searches only after the last answer cue. A *byte-pinned reader* is the frozen reader implementation identified by its file hash; `split_choices_echo` is its single forward scan that removes only a leading model-generated option enumeration, with no fallback. A *cap* is the maximum generated-token budget. All reported p -values identify the reader they were computed under and are shown in mathematical notation. Appendix Table 12 records each primary run’s data root, generation cap, reader, sample count, and seed, with the caps listed as separate runs rather than merged into a decision cell. The *oracle selector* holds retrieved evidence fixed across arms. *At-cap* is a truncation proxy, while *floor-gated* marks a cell too weak to support the registered claim. The *conditional con-*

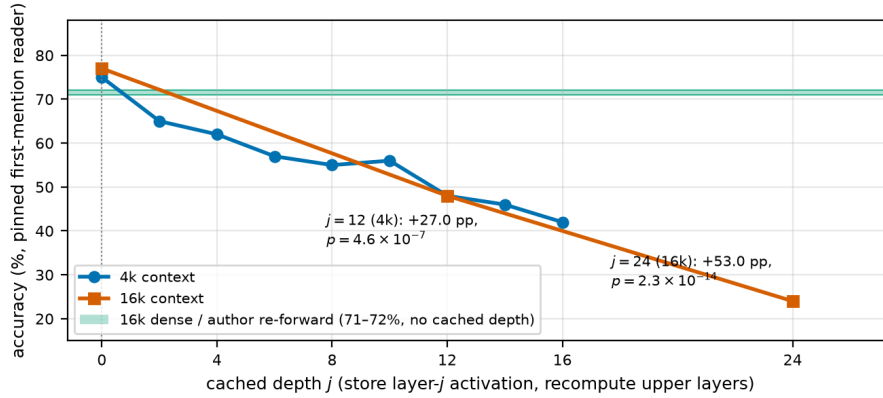


Figure 2: **Complete depth-partition curve** (preregistered sweep; pinned reader, oracle selector, $n=100$ /bucket/arm). Accuracy trends downward with cached depth j at both 4k (circles) and 16k (squares), and every tested point from $j=2$ onward is a significant paired loss against $j=0$; the trend is not asserted to be strictly monotone (the 1 pp reversal at $j=8 \rightarrow 10$ is within noise at $n=100$). The shaded band is the 16k dense and full-depth re-forward (no retrieval) baselines, which have no cached depth and therefore no abscissa here; neither differs significantly from $j=0$ at $n=100$. All tests are pointwise and uncorrected; the family-wise reading is in Table 5.

vention range is the min–max over the six displayed reader/cap cells; the *readout-identity envelope* instead assigns unresolved answers adversarially. Neither is an unconditional confidence interval.

Five-arm cross-trial pairing audit. Pairing for the five-arm decomposition is by input content, not by row or question string: `sample_id = sha256(context+prompt+question)[:16]`. This is the five-arm decomposition pairing (arms A/D/R/C/E), distinct from the decisive single-supporting-fact three-arm pairing reported under Appendix D (which cross-checks the pinned $j=0/j=12$ /re-forward qa1 cells). At each length, A/D/R/C (Trial 1895844) and the E-only repair (Trial 1896955) emit the same ordered sequence of 100 unique IDs; qa1 has only four question strings and six targets, so those strings are not pairing keys. The failed first E attempt stopped in argument parsing before creating an E directory or any scorable output, so the repair run was not selected after seeing an E result. An independent cross-trial audit of same-arm, same-code executions finds only 3 endpoint flips in 500 pairs (0.6%; one-sided 95% Clopper–Pearson upper bound 1.54% (Clopper & Pearson, 1934), with $|\Delta_{\text{acc}}| \leq 0.01$ per unit). By contrast, overturning the E–A exact test requires 8 directed flips at 4k and 11 at 16k, 5–7 $\times$ that worst-case noise bound. Raw strings change on 51–69% of cross-trial rows and therefore have no discriminating power; inference uses only the official-ruler endpoint and its paired b/c counts.¹

Reproducibility statement. Appendix H records the frozen Qwen3-8B/bfloat16 software stack, checkpoint and tokenizer hash prefixes, readers, selectors, seeds, generation caps, hardware identities, and the limits of pairing verification. The ethics and generative-figure disclosure follows the reproducibility record.

¹The pairing, endpoint-noise, and tipping-point figures were independently recomputed from the frozen trial trees in the third-party audit records `r82_pairing_and_crosstrial.txt`, `r82_ENDPOINT_CROSSTRIAL_NOISE.txt`, and `r82_noise_vs_margin_tipping.txt`.

ctx	$j=0$	deeper read	$j=12$	Δ_{j0-j}	exact p
4k	75.0	$j=2$	65.0	+10.0 pp	2.1×10^{-2}
4k	75.0	$j=4$	62.0	+13.0 pp	9.8×10^{-4}
4k	75.0	$j=6$	57.0	+18.0 pp	1.2×10^{-4}
4k	75.0	$j=8$	55.0	+20.0 pp	1.1×10^{-5}
4k	75.0	$j=10$	56.0	+19.0 pp	6.6×10^{-5}
4k	75.0	$j=12$	48.0	+27.0 pp	4.6×10^{-7}
4k	75.0	$j=14$	46.0	+29.0 pp	1.1×10^{-6}
4k	75.0	$j=16$	42.0	+33.0 pp	1.0×10^{-7}
16k	77.0	$j=12$	48.0	+29.0 pp	1.3×10^{-7}
16k	77.0	$j=24$	24.0	+53.0 pp	2.3×10^{-14}
16k dense [†]	–	–	71.0	–	0.43
16k re-forward [†]	–	–	72.0	–	0.50

Same-run wall clock (s, $n=100$), recorded arms only						
arm	4k $j=12$	4k $j=16$	16k $j=12$	16k $j=24$	16k dense	16k re-forward
seconds	504.3	501.7	511.6	606.1	734.3	750.6

Table 5: **Complete depth-partition curve (9+3 points, preregistered sweep) with the same-run wall clock.** Accuracy (%) under the pinned first-mention reader ($n=100$ /bucket/arm, oracle selector); paired exact McNemar against the $j=0$ arm of the same length, which shares the selected raw chunks (retrieval fixed) and the content-hash-verified paired subset (Appendix D). Every tested point from $j=2$ onward is a significant paired loss against $j=0$ on the same samples; the trend in j is downward overall but we claim no strict monotonicity (the single 1 pp reversal at $j=8 \rightarrow 10$ at 4k is within noise at $n=100$, and each p compares a depth to $j=0$, not to its predecessor). The p values are pointwise: under Bonferroni over the eight 4k comparisons the shallowest $j=2$ cell ($p=2.1 \times 10^{-2}$) does not survive $0.05/8=6.25 \times 10^{-3}$, so “every point from $j=2$ ” is pointwise, not family-wise. The load-bearing $j=12$ point is +27.0 pp at 4k; at 16k the $j=12/j=24$ losses are +29.0/+53.0 pp. [†]dense is stock full-context generation (no CoMem) and re-forward the authors’ full-depth re-forward without retrieval (not an implementation of the published KV-Direct method): at 16k neither differs significantly from $j=0$ at $n=100$ ($p=0.43/0.50$) — a non-rejection, not equivalence — and their ≤ 6 pp deficit is the registered bound on retrieval-side differences. The lower block is the wall clock of the same runs, so quality and cost are matched operation points: the matched $j=0$ references are 465.6 s at 4k and 528.5 s at 16k, and seconds were not retained for the intermediate 4k depths $j=2$ –10. Per-depth peak memory was not recorded (Appendix J); data roots are in the protocol matrix (Appendix Table 12).

D PAIRING PROVENANCE ON THE CONTENT-HASH-VERIFIED SUBSET

The preregistered sweep CSVs never emitted a separate `context.sha256` column (see the CSV column disclosure in Appendix H). They do emit `sample_id`, which the harness defines as the 16-hex prefix of `sha256` over the full formatted model input, i.e. over the concatenation of context, prompt, and question (`eval/babilong.py` line 242). That column is therefore itself a content hash, and we re-ran the preregistered content-level join on it for every decisive read-depth cell rather than leaving the field unverified.

Table 6 reports the outcome. In all 16 recomputed cells each arm contributes 100 rows carrying 100 distinct content hashes, the two arms share all 100 hashes (no arm-only and no duplicated hash), and the stored question string agrees on every shared hash. Recomputing each published cell on that verified subset reproduces the printed accuracies, gaps, discordant b/c counts, and exact McNemar p values *exactly*: every comparison in the recompute record is a match, so no number in this paper changes. What changes is the provenance status — the decisive $j=0$ versus $j=12$ cells are now certified to sit on a content-hash-verified paired subset rather than on stored row order alone. One limitation remains and is not repaired by this recompute: the full 64-hex digest of the raw document text was never retained as an independent artifact, so the join is over the formatted model input, not over a separately stored document digest.

cell group	cells	rows	hashes	shared	arm-only	q-mism.
mnt160 read-depth sweep (0k–8k)	5	100	100	100	0	0
mnt512 decisive cell	1	100	100	100	0	0
depth sweep (4k $j=2-16$; 16k $j=12, 24$)	10	100	100	100	0	0

Table 6: **Content-hash join for every decisive read-depth cell.** The join key is the harness `sample_id`, a 16-hex `sha256` prefix over the full formatted model input. *rows* and *hashes* are per arm (rows contributed and distinct content hashes); *arm-only* counts hashes present in one arm and absent in the other; *q-mism.* counts shared hashes whose stored question string disagrees between arms. No hash is duplicated within an arm in any cell. Every cell is a complete 100/100 overlap, so all read-depth values in Tables 5 and 2 are computed on the verified paired subset, and the recomputed values match the published ones cell by cell.

Pairing-free floor for the load-bearing cell. Because a reviewer cannot audit the retired raw contexts directly, we also report the estimate that survives with *no* pairing assumption at all. Treating the 4k pinned cell as two independent binomial arms (75/100 versus 48/100) gives a +27.0 pp difference with pooled $z = 3.924$ and two-sided $p = 8.7 \times 10^{-5}$, and a Newcombe square-and-add interval of $[+13.6, +39.1]$ pp computed with the pairing correlation set to zero. That interval is deliberately wider than the paired constructions in Table 16, which use the observed discordance: the paired test is the efficient estimator, and the unpaired two-proportion result is the assumption-free bound. Both exclude zero, so the direction does not depend on the pairing evidence. The same construction applied to the mnt512 decisive cell gives $[+10.7, +36.1]$ pp (pinned) and $[+10.2, +34.8]$ pp (stripped), again wider than and consistent with the paired intervals.

E RELOCATED SERVING RECORD AND FIGURES

Table 7 is the complete measured-quantity record for the serving estimand, and Figures 3 and 4 are the serving overview and the read-depth length profile. All three are supporting views of Section 5.3 and Section 5.1; the load-bearing read-depth estimates remain in Table 2.

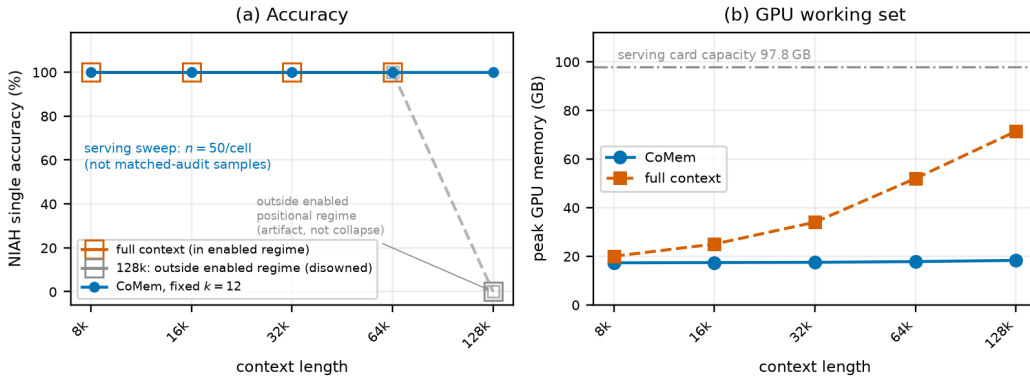


Figure 3: **Fixed- k preserves the short-read operating point while full-context cost grows.** Panel (a) is the separate `niah.single` serving sweep ($n=50/\text{cell}$), not Table 11’s `niah.single_3`. The 128k full-context point is greyed and annotated *outside enabled positional regime*: it lies outside this checkpoint’s enabled positional range and is disowned as a positional artifact, not a full-attention-collapse finding. Panel (b) adds the 97.8 GB serving-card capacity line, below which full attention remains feasible at 128k (71.4 GB). The former read-time panel is now the recorded read/prefill rows of Table 7, because three endpoints are not a sweep. The depth-axis quality cost is plotted in Figure 2 and Table 5 instead; these protocols are not one paired Pareto sweep.

Block	Arm / estimand	Protocol	Recorded result
128k serving	Full context	all raw tokens; all layers	72.7 s read; 71.4 GB
	Full context at 8k	shortest recorded	1.28 s read
		full-context read	
	Matched $j = 0$ timing	selected raw tokens; same k	0.94 s read
	CoMem $j = 12$	selected h_{12} ; same k	0.68 s read; 18.29 GB
	Balanced break-even	$W = 6.65$ s, $G = 1$; decode excluded	$Q^* = 25.58$ reads/document (≈ 26)
E1 matched memory	$j = 0$	read length 6,657; median of 3	18.2 GB
	$j = 12$	same read/hardware; median of 3	17.9 GB
	Depth-attributed saving	$(12/36)B_{KV} read_len$	0.3 GB (0.2–0.4 GB)
Separate quality	RULER macro, <i>estimand mixture</i>	unweighted mean over 15 cells: 3 fractional <code>string_match_all</code> recall, 12 strict-binary	99.19 $\rightarrow$ 96.07; gap 3.12 pp
	RULER strict-binary	1,500 item pairs, $(c - b)/n$	82/1500 = 5.47 pp
	pooled drop		
	RULER <code>niah_single</code>	serving sweep, $n=50$ /cell, 8k–128k	100% at every in-regime length
	fixed read	unpaired, $n = 500$ per arm; $k \in \{4, 6\}$	97.2 $\rightarrow$ 69.0; 28.2 pp
	LongEval		[23.9, 32.5]; $z = 11.9$

Table 7: **Measured quantities split by operating point.** The serving block is a whole-system retrieval comparison of recorded read/prefill endpoints; E1 is the matched memory-only depth isolation, scoped to `read_len=6,657`, and establishes no end-to-end or general memory benefit. The RULER macro row is explicitly an *estimand mixture*: an unweighted mean over 15 (task, length) cells of which three report fractional `string_match_all` recall and the rest strict-binary correctness, so it is not a single-estimand accuracy. The comparable strict-binary quantity is the pooled 5.47 pp drop; the item-level pooled exact test over 1,500 clustered pairs is descriptive only and appears with its per-cell breakdown in Appendix Table 11. The 100% row is the `niah_single` serving sweep, a different subtask and sample set from that appendix table. LongEval is unpaired with a different selector budget, so it is not a same-retrieval depth isolation and its 95% Wald interval assumes independent binomial arms. Timing dispersion was not retained; serving timings are medians of three.

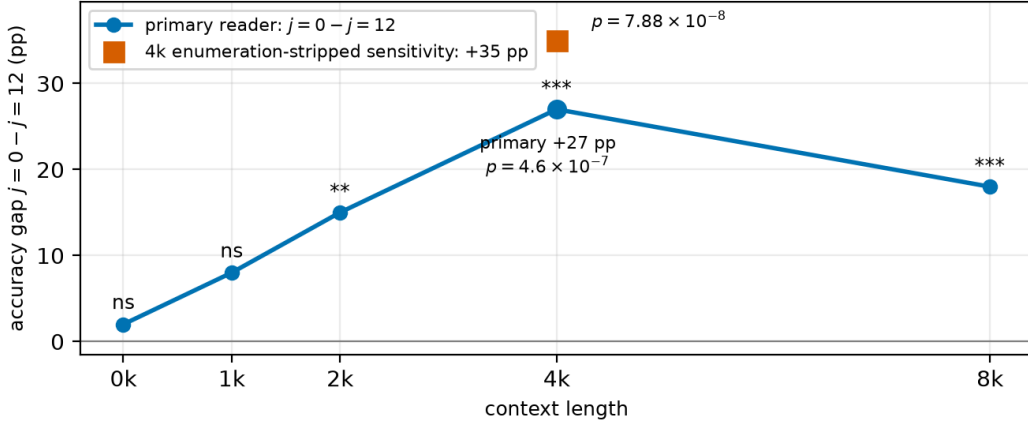


Figure 4: Primary-reader $j=0 - j=12$ qa1 gaps over context length ($n=100$ /bucket/arm, oracle selector). The square is the enumeration-stripped sensitivity at the 4k decision cell. The curve is not monotone; the load-bearing result is the pinned *direction* and the +27.0/+24.0 pp point pair on the same samples at the two caps. The conditional convention range is in Table 8; neither is a claim of length-robust growth. Symbols encode pointwise exact McNemar tests: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, and “ns” $p \geq 0.05$; no multiplicity correction is applied in this panel.

F EXTENDED LIMITATIONS

Selection and uncertainty. The archived best- k sweeps lack a validation/test separation, so we report fixed $k = 12$. Primary RULER and qa1 cells have $n = 50$ and $n = 100$ under one checkpoint and deterministic data seed. The qa1 oracle selector isolates depth but leaves its interaction with deployment BM25 unmeasured. Overlapping half-splits diagnose resampling stability, not independent replication; our intervals do not cover checkpoint, model-family, or implementation variability. No column literally named `context_sha256` was emitted, so the recomputed join uses the emitted `sample_id` content hash and is therefore over the formatted model input rather than over an independently stored 64-hex document digest, which was never retained (Appendix D); for the load-bearing cell we additionally report the pairing-free two-proportion floor.

Scorer resolution of the five-arm deck. The official scorer collapses roughly six in ten raw-generation differences onto a single label, so the 0/100 endpoint agreement between arms (e.g. E and C) is indistinguishability *at label resolution*, not identity or inertness of a deleted factor in the forward pass. We therefore report the generation-layer distinguishability counts alongside the endpoint discordance; the null is a scorer-resolution statement only.

Composition boundary and the archived read adapter. The qa2 probe is not load-bearing: one cell is parser-invariant and non-floor; the rest are negative-control, format-artifact, or floor-gated cells. Appendix Table 9 reports them, together with the registered generation-cap sensitivity runs whose constant direction is consistent with the gap not being created by generation budget alone; no length-robust claim follows, and other format effects are not ruled out. A self-distilled read adapter appears in this workspace only as an archived configuration (Appendix Table 14): it is not evaluated here and carries no accuracy claim in this paper.

G SECONDARY AND SENSITIVITY RESULTS

Full read-depth convention matrix. Table 8 preserves all six displayed reader/cap cells of the decisive qa1 arm, including the two *missing-excluded* rows with their independent $n = 59/80$ and the conditional convention range. The canonical main-text view (Table 2) promotes only the two pinned decision rows, so this table remains the complete record of the convention sensitivity.

Reading the blank cells of the canonical read-depth table. The *re-fwd.* column of Table 2 is the authors’ *full-depth re-forward without retrieval* control arm — CoMem with $j=0$ and no depth cache — and *not* an implementation of the published KV-Direct method (Qasim et al. (2026)). A full-context arm was run only at 16k (Table 5) and is format-incompatible at the 4k decision cell (Table 9), so it is not a comparison cell there. At 4k the re-forward control is 73% versus $j=0$ 75%; no paired test was run at 4k, so we report no “coincides” claim, and the 16k control comparison is the depth-curve table’s *re-forward* row. The mnt512 rows carry no *re-fwd.* cell because that control was not executed at the 512 cap; the dash records an absent run, not a null result.

data root	missing-answer convention	$j=12$	$j=0$	n	Δ_{j0-j12}	exact p	b/c
mnt160	pinned first-mention	48.00%	75.00%	100	+27.0 pp	4.6×10^{-7}	2/29
mnt160	stripped, missing wrong	40.00%	75.00%	100	+35.0 pp [†]	7.9×10^{-8}	5/40
mnt160	stripped, missing excluded	64.41%	86.44%	59	+22.0 pp	4.4×10^{-3}	3/16
mnt512	pinned first-mention	52.00%	76.00%	100	+24.0 pp	8.4×10^{-6}	3/27
mnt512	stripped, missing wrong	57.00%	80.00%	100	+23.0 pp	1.17×10^{-4}	6/29
mnt512	stripped, missing excluded	70.00%	86.25%	80	+16.2 pp	1.1×10^{-2}	5/18

Table 8: The matched read-depth direction survives both generation caps and all three read-out conventions. The mnt160/mnt512 roots reuse the same 100 samples at caps 160/512, not independent datasets; for missing-excluded rows accuracy is over the retained pairs answered under both protocols (independent $n = 59/80$). The zero-free-parameter pinned rows are the primary magnitude and are the two rows promoted to the canonical main-text view (Table 2): their exact paired 95% CIs on the risk difference are $[+17.7, +30.5]$ pp (mnt160) and $[+14.1, +28.7]$ pp (mnt512), both excluding zero. The $[16.2, 35.0]$ pp range is only the conditional min/max over the six cells; both endpoints use the stripped reader, so a wider strip rule can reverse the sign, and the pinned rows do *not* inherit it. A separate answer-anchored mnt512 reader gives a null cell ($p=0.0649$, conditional CI crossing zero) and is not folded into this six-cell range. Stripped digits use canonical `split_choices_echo`, not an independent rescoring. All six cells sit on the content-hash-verified paired subset (Appendix D). The interval construction and its cross-checks are in Appendix J.1; Table 10 reports headroom and parser diagnostics.

Composition probe. Table 9 reports every registered qa2 cell and classifies its evidentiary role. Only the parser-invariant, non-floor cell is descriptive; the remaining rows are a negative control, a format artifact, or floor-gated.

(a) Context composition at the 160-token cap (registered scoring reader)						
ctx	$j = 12$	$j = 0$	re-forward	Δ	exact p	interpretation
0k	56	57	57	+1	1.00	negative control
1k	52	54	55	+2	0.839	negative control
2k	32	43	53	+11	1.9×10^{-2}	descriptive
4k	30	44	49	+14	4.3×10^{-3}	not-evaluable
8k	23	38	53	+15	8.1×10^{-3}	floor-gated

(b) Registered generation-cap sensitivity at 4k						
cap	$j = 12$	$j = 0$	re-forward	Δ	exact p	interpretation
20	12	28	—	+16	—	budget sensitivity
64	11	27	—	+16	—	budget sensitivity
160	11	27	—	+16	0.00154	budget sensitivity

Table 9: **BABILong qa2 composition and cap-sensitivity probes.** Panel (a) is scored with the same registered parser family as the qa1 decisive table (reader and data root are consolidated in the protocol matrix, Table 12). The registered overall verdict here is *not evaluable*: both 4k and 8k separate, but the floor gate fails in a load bucket ($j=12$ at 8k is $23 < k^*=24$). The probe therefore does not support a length-robust read-depth claim and is not load-bearing. The *re-forward* column is the authors’ full-depth $j=0$ control without retrieval, not an implementation of a published method; no full-context arm was run on qa2. Panel (b) keeps the three registered legacy cap runs separate. At cap 160 the exact paired result is $p = 0.00154$, with disjoint 95% arm intervals $[6.3, 18.6]\%$ and $[19.3, 36.4]\%$.

The fixed-read quality cost is task-dependent. The $j=0-j=12$ gap is not length-uniform and the fixed top- k read is task-dependent: on the matching-style RULER probes (Table 11) the fixed read retains 90–100% recall, and on the single-supporting-fact qa1 sweep (Table 5) deeper reads degrade with cached depth. The multi-hop qa2 composition probe (Table 9) is *not* load-bearing: under the registered adjudicator its overall verdict is not evaluable (the $j=12$ load arm falls below the $\geq k^*=24$ floor at 8k), so no clean “ $j=0$ loses X pp” quantification is registered for composition. Consistent with the qa2 panel’s negative-control and floor-gated cells and with the author re-forward versus dense disclosure above, we do not claim that a fixed read preserves accuracy on compositional or multi-hop queries; the main-text quality claim is scoped to single-supporting-fact, needle-style tasks.

Exploratory difficulty partition. Using the pinned reader and the registered ‘split-choices-echo’ rule (its definition and exact tokenization are consolidated in the protocol matrix, Table 12), we partitioned questions by shallow-read success without new model execution. In the *hard* stratum (under the pinned readout), the $j = 0$ re-forward recovers roughly 55–56% from near-zero $j = 12$ accuracy; the already-correct partition has no positive gap. The direction survives all three missing-answer conventions. This post-hoc, small-stratum diagnostic is not a new scorer or a load-bearing magnitude: it is kept separate from both the convention band and the readout-identity envelope carried in Table 8, and is disclosed only in the appendix.

Read-out invariance diagnostics. Table 10 preserves the complete numerical checks summarized in Section 5.1. They constrain interpretation but do not overturn the two-root directional result in Table 8.

check	registered observation	interpretation
Effective at-cap proxy	At both caps, outputs exactly at the maximum whitespace-token length are 1% ($j = 12$) versus 18% ($j = 0$).	The better arm is cut more, so truncation would compress rather than create $j_0 > j_{12}$.
Legacy punctuation control	Outputs lacking terminal punctuation: 89% ($j=12$), 90% ($j=0$), and 100% (full context).	Absorbing style proxy; not used as the truncation gate or as load-bearing evidence.
Prefix-pinned envelope	At cap 160, the $j=0/j=12$ arm ranges are [75, 79]/[48, 62] pp; at cap 512 they are [76, 79]/[52, 59] pp. Adversarial endpoint pairing gives gap envelopes [+13, +31] and [+17, +27] pp, respectively.	Both envelopes exclude zero; this preserves sign, not a point magnitude.
Stripped reader and headroom	At mnt512, un-stripped $j=12=52\%$ versus $j=0=76\%$ (+24.0 pp) becomes, under the stripped reader, $j=12=57\%$ versus $j=0=80\%$ (+23 pp, $b=6/c=29$, exact $p=1.17 \times 10^{-4}$); the strip rule is <code>split_choices_echo</code> : the model's self-generated option list is removed by a single forward scan before answering, with no fallback, applied identically to both arms. Only 6 of the original $c=27$ discordant rows can reverse with unlimited generation, leaving at least +18 pp. The second-root stripped result is in Table 8.	Direction survives; magnitude is not clean.
Answer-anchored reader	mnt160: 33% versus 51%, +18.0 pp, $b=19/c=37$, exact $p=0.0222$. At mnt512: 43% versus 57% ($n=100$), +14.0 pp, $b=18/c=32$, exact $p=0.0649$; exact conditional 95% CI [-0.81, +27.08] pp (Wald [+0.1, +27.9], paired-McNemar [+0.4, +27.6]), 46% power at $n=100$ and $n \approx 210$ for 80% power.	Answer-anchored sign agrees on the same samples at both caps; the 512-cap result is underpowered and not confirmation.
Full-context parser	Anchored accuracy is 11% versus the 16.7% chance floor ($p=0.078$), and 3.0% on the second root ($p=1.8 \times 10^{-5}$); only 13% of outputs contain an answer-adjacent label, versus pinned readouts of 74.0%/73.0%.	Format incompatibility, not an effect estimate.

Table 10: **At-cap and answer-anchored checks preserve direction but not a point magnitude.** The answer-anchored reader scans forward at most 120 characters from the last *answer* cue with no fallback; the exact implementation of `anchored_label` (`split_choices_echo` and `anchored_label` both) and their pinned hashes are consolidated in the protocol matrix (Appendix Table 12). These sensitivity checks retain the direction of the primary result but do not establish readout invariance or a clean effect magnitude.

G.1 TRUNCATION ENVELOPE DERIVATION

Table 10 collects the prefix-pinned arm ranges and the resulting two-cap envelopes. To bound readout identity without treating it as known, the registered worst-case construction assigns each unresolved missing-answer row adversarially per arm: a lower endpoint pairs the smallest $j=0$ reading with the largest $j=12$ reading, and the upper endpoint uses the reverse extremes. Neither envelope contains zero, so this construction preserves the sign of the pinned result. Both envelopes use only the qa1 decisive samples and reader; they are not point estimates and do not replace the retained convention band in Table 8.

Table 10’s at-least-+18 pp headroom result is a different estimand from the envelope above: rather than assigning the unresolved rows adversarially, it conditions on a particular recovery rule (the post-hoc stripped reader at cap 512) and reports the smallest headroom consistent with that rule, leaving at least +18 pp. The per-row discordant counts and the resulting bound are given in Table 10. The two constructions therefore have different scopes, but both agree in sign with the pinned direction.

Secondary diagnostic boundary. Archived cross-pack, iterative-retrieval, and 80-layer MoE diagnostics are not used for the headline. Their code paths survive, but the bundle does not pin a

common validation protocol or every run-specific training field; we avoid a second, incomparable number grid.

cell	arm $j=0$	arm $j=12$	Δ (pp)	$j=0$ err.	$j=12$ err.
niah_single_3/8k	100.0	100.0	0.0	0	0
niah_single_3/16k ^M	97.0	91.0	6.0	3	9
niah_single_3/32k	97.0	97.0	0.0	3	3
niah_single_3/64k	99.0	98.0	1.0	1	2
niah_single_3/128k	98.0	98.0	0.0	2	2
niah_multikey_1/8k	100.0	94.0	6.0	0	6
niah_multikey_1/16k	100.0	91.0	9.0	0	9
niah_multikey_1/32k	100.0	99.0	1.0	0	1
niah_multikey_1/64k	97.0	90.0	7.0	3	10
niah_multikey_1/128k	100.0	93.0	7.0	0	7
variable_tracking/8k	99.8	96.2	3.6	–	–
variable_tracking/16k	100.0	98.0	2.0	–	–
variable_tracking/32k	100.0	98.2	1.8	–	–
variable_tracking/64k	100.0	98.6	1.4	–	–
variable_tracking/128k	100.0	99.0	1.0	–	–

Table 11: **Matched RULER losses are concentrated in a few cells.** Arm $j = 0$ is the full-depth re-forward and $j = 12$ the cached read, $n = 100/\text{cell}$. NIAH rows are strict-binary. The variable_tracking rows instead show fractional string_match_all recall and therefore carry no per-cell integer error counts. For the pooled audit they are binarized only by `int(recall >= 1.0)`; exact McNemar operates on these paired full-match labels and is excluded from any per-cell identity lacking integer errors. The macro partial-credit drop is 3.12 pp; the strict-binary pooled drop is $(c-b)/n = 82/1500 = 5.47$ pp, i.e. 13 strict-binary errors at $j=0$ against 95 at $j=12$ over the pooled 1,500 item pairs (the variable_tracking contribution enters only through the binarized labels, which is why the listed per-cell integer columns sum to fewer errors). Because the 1,500 items are clustered in 15 (task,length) cells, the pooled McNemar $p \approx 8.8 \times 10^{-24}$ over 84 discordances is reported only as *descriptive*. No cell-clustered inferential test is reported, so the pooled test does not carry a family-wise significance claim. Under the paper’s convention $b=1/c=83$: b counts $j = 12$ -correct/ $j = 0$ -wrong and c the reverse. Superscript M marks a matched study cell, not Figure 3’s separate niah_single serving-sweep sample set.

H REPRODUCIBILITY AND PROVENANCE

(a) Run identity and sample accounting

run	data root	cap	n	seed
qa1 decisive	mnt160	160	100/bucket/arm at 0, 1, 2, 4, 8k	preregistered
qa1 invariance	mnt512	512	same 100 samples/arm at 4k	preregistered
qa2 budget	budget root	20	100/arm at 4k	preregistered
qa2 budget	budget root	64	100/arm at 4k	preregistered
qa2 budget	budget root	160	100/arm at 4k	preregistered
RULER serving	released	512	50 per cell/arm	42
	RULER			
RULER matched	15 cells	512	$2 \times 50 = 100$ per cell/arm	42
LongEval	released lines	512	50 per length, task, arm	1234
H20 serving	serving record	–	median of 3	42
E1 matched	two 128k trial	–	2 trials $\times$ 3 repeats/arm	fixed
memory	roots			
E2 five-arm	merged trial	160	100 per length/arm at 4k, 16k	fixed
	roots			

(b) Reader, metric, and selector definitions

run	reader / metric	selector
qa1 decisive	pinned / stripped / anchored	oracle, $k = 12$
qa1 invariance	pinned / stripped / anchored	oracle, $k = 12$
qa2 budget	first mention	oracle, $k = 12$
RULER serving	string-match recall	fixed read
RULER matched	recall / strict full match	oracle / retrieval-fixed
LongEval	exact line identifier	BM25 $k \in \{4, 6\}$
H20 serving	CUDA peak / prefill time	matched endpoints
E1 matched	CUDA peak at <code>read.len=6,657</code>	fixed $k = 12$
memory		
E2 five-arm	official <code>compare_answers</code> qa1 endpoint	oracle, $k = 12$

(c) Artifact and pin identifiers (stable hashes)

artifact	identifier
Depth-partition sweep data	<code>depth_sweep_full.v2.20260823.201948</code>
qa1 five-bucket readout	<code>readdepth.budget.20260821.083604</code>
qa2 amend3 verdict	<code>readdepth.budget.20260821.053103</code>
Pinned paired-CI input	<code>qa1.4k.pinned.paired.CIs</code>
qa1 decision / readout prereg	SHA-256 pins <code>2e18f293 / 87fef65b</code>
Reader-label implementation	<code>anchored.label / split.choices_echo</code> (pinned SHA in code snapshot)
E1 memory evidence	<code>matched.j0_peak.20260824-{210326, 212122}</code>
E2 merged scores	<code>e2_FULLL.5arm.e2_dual_step.20260825.001307</code> ; Code snapshots: A/D/R/C under <code>572907d7</code> ; E under <code>7f4aa8aa</code> (the choices-only <code>depth_only</code> repair; the cross-trial harness provenance paragraph in this appendix carries the side-effect certificate)
Independent E2 audit	<code>r82.pairing.and.crosstrial</code> ; <code>r82.ENDPOINT.CROSSTRIAL.NOISE</code> ; <code>r82.noise.vs.margin.tipping</code>

Table 12: **Protocol matrix keeps generation caps and estimands separate.** Panel (a) records run identity and sample accounting; panel (b) separates reader/metric definitions from selectors; panel (c) consolidates the stable artifact and hash identifiers (the registered data roots, decision verdicts, CI input, preregistration pins, and reader-label implementations), which are referenced by name elsewhere instead of as mutable local paths. The `mnt160` and `mnt512` rows reuse the same qa1 samples at two caps, whereas the three qa2 cap rows are distinct executions. RULER fractional recall and strict full-match labels are likewise not merged into one decision cell.

Code, data, and artifact availability. Write/read, selection, and RULER evaluation live in the modules `comem.model`, `comem.selectors`, and `eval.ruler`; all three quantitative figures are produced by deterministic scripts in the manuscript build tree, and the exact running inputs are enumerated in the protocol matrix (Table 12). An anonymized artifact accompanies review rather than waiting for acceptance: it contains the write, read, selector, scorer, and evaluation drivers, the frozen readers, the preregistration documents, the per-arm scored CSVs behind every table in this paper, and the figure scripts, with all account-bound path identifiers removed. Hash convention: every identifier printed in this paper is a truncated SHA-256 prefix, and the artifact manifest lists the full 64-hex digest for each file it ships, so a reviewer can extend any prefix below to a complete digest. Two identifiers cannot be extended and we do not pad them: the tokenizer artifact survives in the run log only at six hex characters (5a7303), and the pre-repair 572907d7 driver blob is not recoverable from history, so its difference against the repaired driver is archived as a diff rather than reproduced.

Cross-trial harness provenance (E2 five-arm deck). The five-arm decomposition merges two trials: A/D/R/C come from Trial 1895844 and the depth-only arm E from the repair trial 1896955 (different harness snapshots: `eval/babilong.py` 572907d7 and 7f4aa8aa respectively). `comem/model.py` is bit-identical across both (0e7ef9e6); the only `babilong.py` difference is the R243 fix adding `depth_only` to the `--ablation` option `choices` list (archived as `docs/C2_R243_babilong_depth_only_diff.txt`). This is a driver CLI-whitelist edit: `args.ablation` is consumed at a single point (`babilong.py:231`) and the choices are bare string literals with no reflection, so arms A/D/R/C (which already accept `none/relocation_only/chunk_indep_only`) have no reachable behavioral difference across the two snapshots, and the `sample_id` (`=sha256(input+prompt+question)[:16]`) sequences are bit-identical between the two trials ($n=100$ each length). The reported scorer (`babilong.metrics.compare_answers`, pinned f77c1398) operates downstream of generation and does not control this gap; the static whitelist argument above is the side-effect certificate. A caveat is that the original 572907d7 blob is not recoverable from git, so the diff is archived, not reproduced from history.

Pairing-identity verification. For the historical decisive qa1 cells, a column literally named `context_sha256` was never emitted, and earlier versions of this work therefore rested pairing on the registered per-arm `sample_id` sequences and the deterministic oracle selector alone. That gap is now closed one level further: `sample_id` is itself the 16-hex sha256 prefix of the full formatted model input, so the preregistered content-level join was recomputed on it for every decisive cell, with a complete 100/100 overlap and no orphaned, duplicated, or question-mismatched context (Appendix D). Pairing therefore no longer rests on row order, and it never rested on the repeated question string (which is not unique). What we still do not claim is a join against a separately stored 64-hex digest of the raw document text, which was not retained. This historical three-arm check is distinct from, and complemented by, the five-arm decomposition pairing described in Appendix C. E2 is different by construction: its key is `sha256(context+prompt+question)[:16]`, and the independently checked A/D/R/C/E files carry the same ordered 100 unique keys at each length. No paired claim is made across 4k and 16k, which are different length cells.

Frozen stack and preregistration. All primary model reads use Qwen3-8B in bfloat16 with 36 transformer layers, PyTorch 2.11.0+cu129, Transformers 5.14.1, Accelerate 1.14.0, and Datasets 5.0.1. The checkpoint config and tokenizer artifacts are pinned by SHA-256 prefixes bc058b40 and 5a7303, respectively. The qa1 reader and decision rules, the 512-cap readout-invariance gate, and their SHA-256 pins are consolidated in the protocol matrix (Appendix Table 12). Both preregistrations went through documented revisions: the qa1 decision rules went through three documented revisions (Table 13), each frozen before the qa1@4k decision cell was inspected. The pack uses one BOS sink, selected chunks in original-document order, fresh contiguous positions, and the query last. Post-hoc selection note: among the valid (non-artifact) pinned magnitude cells {2k, 4k, 8k}, the 4k decision cell carries the largest significant gap (+27 pp; Table 2). The preregistered anchor was documented to move $1k \rightarrow 4k$; the headline is therefore best read as a selected-magnitude negative result rather than a pre-specified effect size. We do not report a selection-adjusted statistic. The full revision timeline (what data was seen before each REVISION) is disclosed in the prereg assets; the pointwise per-cell p values are in Table 2.

Revision	Change
#1	initial qa1 decision rules
#2	anchor moved $1k \rightarrow 4k$; Branch-B third conjunct and decision criterion updated
#3	repaired three adjudicator code defects: sampling-error units; integrity adjudication of target mismatch; acceptance coupling

Table 13: **qa1 preregistration revision timeline.** Each decision rule set was frozen before the qa1@4k decision cell was inspected; what data was seen before each revision is disclosed in the prereg assets.

One-million-token serving point. The recorded 1M context measurement is approximately 25.6 GB, +44% versus 32k, and is indexed (with the other serving claims) in the protocol matrix (Table 12). It is a separate serving measurement, not an interpolation of Figure 3 and unrelated to the read/prefill break-even $Q^*=25.58$ reads/document reported in Section 5 (the coincident digits denote different quantities).

Matched-memory scope and preregistered scale check. At 128k and fixed $k = 12$, E1 records $read_len=6,657$ and median-of-three peaks of 18.2 GB for $j = 0$ and 17.9 GB for $j = 12$ in each of two trials (six of six repeats agree at 0.1 GB print precision). The measured 0.3 GB difference is the read-length-scoped analytic quantity $(12/36)B_{KV} read_len$, not evidence that depth caching is generally free. We preregister a $k = 48$ matched check: because the analytic band scales linearly with read length, it predicts approximately 1.2 GB of depth-attributed saving. That check has not been run and is not a result of this paper.

Self-distillation boundary. The archived report pins the settings in Table 14. Implementation defaults are not substituted for missing run evidence, and the adapter result is not primary evidence.

Setting	Value / status
Corpus	PG19
Cached depth j	32
LoRA rank	32
LoRA scale	64
Target layers	upper layers
Steps	4000
Learning rate, effective batch/tokens, corpus split, checkpoint selection, seed	not pinned
Stopping criterion	step count only

Table 14: **Self-distillation configuration archived by the report.** The pins and the explicitly unpinned settings are listed so no implementation default is silently substituted for missing run evidence.

Algebraic check. The self-test compares a contiguous forward with split/rejoin on identical tokens and positions. It validates layer composition only, not chunk-local writes, selection, relocation, or the empirical $j > 0$ read measured in Section 5.1.

I ETHICS AND AI-ASSISTED FIGURE DISCLOSURE

No human-subject data were collected. Source text and activations carry the privacy and deletion obligations in Section 6. A generative image tool produced the initial architecture raster; its retrieval arrows, position labels, store growth, and fixed-read boundary were iteratively checked and corrected. Both quantitative figures come only from deterministic scripts and registered measurements; generative tools did not create or alter experimental data.

J DEPTH-SWEEP COST AND DATA DISCLOSURE

The same-run wall clock for the depth-partition sweep is merged into Table 5 so that the eight sweep arms are tabulated once instead of twice. All rows are $n = 100$ at the stated context length, scored under the byte-pinned reader; every arm shares the same selected raw chunks (retrieval fixed), so *quality* and *cost* are measured on the same operation point for each arm—these points are not drawn from different operation points. Per-depth peak memory (`peak_memory_gb_qmem`) was *not* recorded for the sweep, so no depth-memory curve exists; the separate E1 0.3 GB point is in Table 7, and the two cards are distinguished in Table 15. Wall-clock seconds were retained only for the arms listed in Table 5 (the intermediate 4k depths $j=2-10$ have no retained seconds), and the run-level wall clock does not support a *time* saving from depth.

CSV column disclosure. The preregistered sweep (data root in Table 12) asked each arm CSV to carry `sample_id`, `context_sha256`, and `robust_correct`. The emitted columns are only `target`, `output`, `question`, and `sample_id` for all 14 arms; a column literally named `context_sha256` is *absent*. The emitted `sample_id` is nevertheless a content hash of the full formatted model input, so the preregistered content-level join was recomputed on it and every decisive cell is a complete 100/100 overlap (Appendix D); what remains unavailable is a separately stored 64-hex digest of the raw document text, and that gap is disclosed rather than reconstructed.

Hardware identity. Two different NVIDIA H20 cards were used (Table 15). A GPU working-set claim must not be read as if a single card produced both; in particular, the serving-study’s 18.29 GB/3.9×/qmem claims do not rely on the depth-sweep card.

Card	Memory	Used by
NVIDIA H20-3e	143771 MiB	depth-partition sweep (recorded in the sweep data root’s <code>gpu.txt</code>)
H20 (separate)	97.8 GB	fixed- k serving study (Section 4) and factorized account (Table 7)

Table 15: **Hardware identity for memory claims.** The two cards are distinct; a GPU working-set figure must not be attributed to both.

J.1 INTERVAL CONSTRUCTIONS FOR THE DECISION CELLS

The pinned rows of Tables 2 and 8 carry the primary magnitude. Their paired risk difference is $(c - b)/n$; the exact conditional interval applies Clopper-Pearson to $c/(b + c)$ and rescales it as $(2p - 1)s/n$, where $s = b + c$ and $p = c/s$ (registered inputs are in Table 12). It conditions on the observed discordant pair count and remains the primary interval; Table 16 also reports the wider Newcombe Method 11 cross-check, which corrects the square-and-add construction by the observed pairing correlation. Earlier versions of this appendix omitted the pairing-free construction on the grounds that it discards the paired structure; we now report it deliberately as the assumption-free floor, because the raw contexts of these historical cells cannot be re-audited by a reviewer. Setting the pairing correlation to zero in the same square-and-add formula gives an interval that uses only the two marginals, so it holds whatever the pairing is, and the accompanying unpaired two-proportion test gives $z = 3.924$, $p = 8.7 \times 10^{-5}$ at the mnt160 decision cell. The ordering of the three rows in Table 16 is the expected one: conditioning on discordance is tightest, the correlation-corrected interval is intermediate, and the pairing-free interval is widest.

Construction	mnt160 (pp)	mnt512 (pp)
Exact conditional (Clopper–Pearson; primary)	[+17.7, +30.5]	[+14.1, +28.7]
Newcombe Method 11 (Newcombe, 1998)	[+17.1, +36.0]	[+14.1, +33.2]
Square-and-add at zero pairing correlation (assumption-free)	[+13.6, +39.1]	[+10.7, +36.1]

Table 16: **95% interval constructions for the pinned decision rows of Tables 2 and 8.** All three constructions exclude zero at both roots. The first two use the paired discordance; the third discards it and is therefore the assumption-free bound quoted in Table 2. The corresponding McNemar odds ratios are 14.5 (mnt160) and 9.0 (mnt512). Applying the pairing-free construction to the stripped mnt512 row gives [+10.2, +34.8] pp, again excluding zero.